$$\frac{dy}{dx} - 2\frac{y}{x} = 3xy^2$$

$$-y^{-2}\frac{dy}{dx} - 2\frac{y^{-1}}{x} = 3x \qquad y^{-1} = t$$

$$-y^{-2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$-\frac{dt}{dx} - 2\frac{t}{x} = 3x$$

$$\frac{dy}{dx} = y^2\frac{dt}{dx}$$

$$\frac{dt}{dx} + \frac{2t}{x} = -3x$$

$$p = \frac{2}{x}$$

$$t = zu$$

$$\frac{dt}{dx} = z\frac{du}{dx} + u\frac{dz}{dx}$$

$$z\frac{du}{dx} + u\frac{dz}{dx} + \frac{2zu}{x} = -3x$$

$$z\left(\frac{du}{dx} + \frac{2u}{x}\right) + \frac{u\,dz}{dx} = -3x$$

$$\frac{du}{dx} + \frac{2u}{x} = 0$$

$$\frac{du}{dx} = -\frac{2u}{x}$$

$$\frac{du}{2u} = -\frac{dx}{x}$$

$$\ln u = \ln x^{-2}$$

$$u = \frac{1}{x^2}$$

$$\frac{dz}{x^2\,dx} = -3x$$

$$\frac{dz}{dx} = -3x^3$$

$$dz = -3x^3\,dx$$

$$z = -\frac{3}{4}x^4 + K$$

$$t = \frac{1}{x^2}\left(-\frac{3}{4}x^4 + K\right)$$

$$t = -\frac{3}{4}x^2 + c$$

$$y^{-1} = -\frac{3}{4}x^2 + c$$

$$y = -\frac{3}{4}x^2 + c$$

$$\frac{1}{y} = -\frac{3}{4}x^2 + c$$

$$y = -\frac{4}{3x^2} + c$$

1) $\frac{dy}{dx} + y = x^2y^2$

$$y^{-2}\frac{dy}{dx} + y^{-1} = x^2$$

$$y^{-1} = t$$

$$-y^{-2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$-\frac{dt}{dx} + t = x^2$$

$$\frac{dt}{dx} - t = -x^2$$

$$P = -1$$

$$\lambda = e^{\int -1dx} = \frac{1}{e^x}$$

$$dt = (t - x^2)dx$$

$$(x^2 - t)dx + dt = 0$$

$$\frac{x^2 - t}{e^x}dx + \frac{dt}{e^x} = 0$$

$$\frac{\partial u}{\partial t} = \frac{dt}{e^x}$$

$$u = \frac{t}{e^x} + g(x)$$

$$\frac{\partial u}{\partial x} = -\frac{t}{e^x} + g'$$

$$\frac{x^2dx}{e^x} = g'$$

$$\int \frac{x^2}{e^x}dx$$

$$\int v\,du = u \cdot v - \int u\,dv \rightarrow x^2e^x - \int x^2e$$

$$u = x^2$$

$$du = 2x\,dx$$

$$dv = e^x dx$$

$$v = e^x$$